ТОМСКИЙ ПОЛИТЕХНИЧЕСКИЙ УНИВЕРСИТЕТ

ИНЖЕНЕРНАЯ ШКОЛА ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ И РОБОТОТЕХНИКИ

РАЗРАБОТКА АЛГОРИТМА ГЕНЕРАЦИИ ТЕКСТОВОГО ОПИСАНИЯ ИЗОБРАЖЕНИЯ НА ОСНОВЕ НЕЙРОСЕТЕВЫХ АРХИТЕКТУР

Выпускная квалификационная работа

Научный руководитель: Спицын В.Г., д.т.н., профессор ОИТ ТПУ

Консультант: Кривошеев Н.А., старший преподаватель ОИТ ТПУ

Выполнил: Ямкин Н. Н.

Группа: 8ВМ22

20.06.2024

ВВЕДЕНИЕ

Актуальность: задача генерации текстовых описаний изображений имеет широкий спектр практических применений.

Вот некоторые из них:

1. Эффективный поиск изображений или видео в больших коллекциях на основе их содержания;
2. Помощь слабовидящим людям (подразумевается использование в том числе голосового помощника);
3. Системы автоматического управления технологическими процессами (промышленные роботы, автономные транспортные средства).

Цель работы: реализовать и исследовать нейросетевые алгоритмы генерации текстового описания изображения.

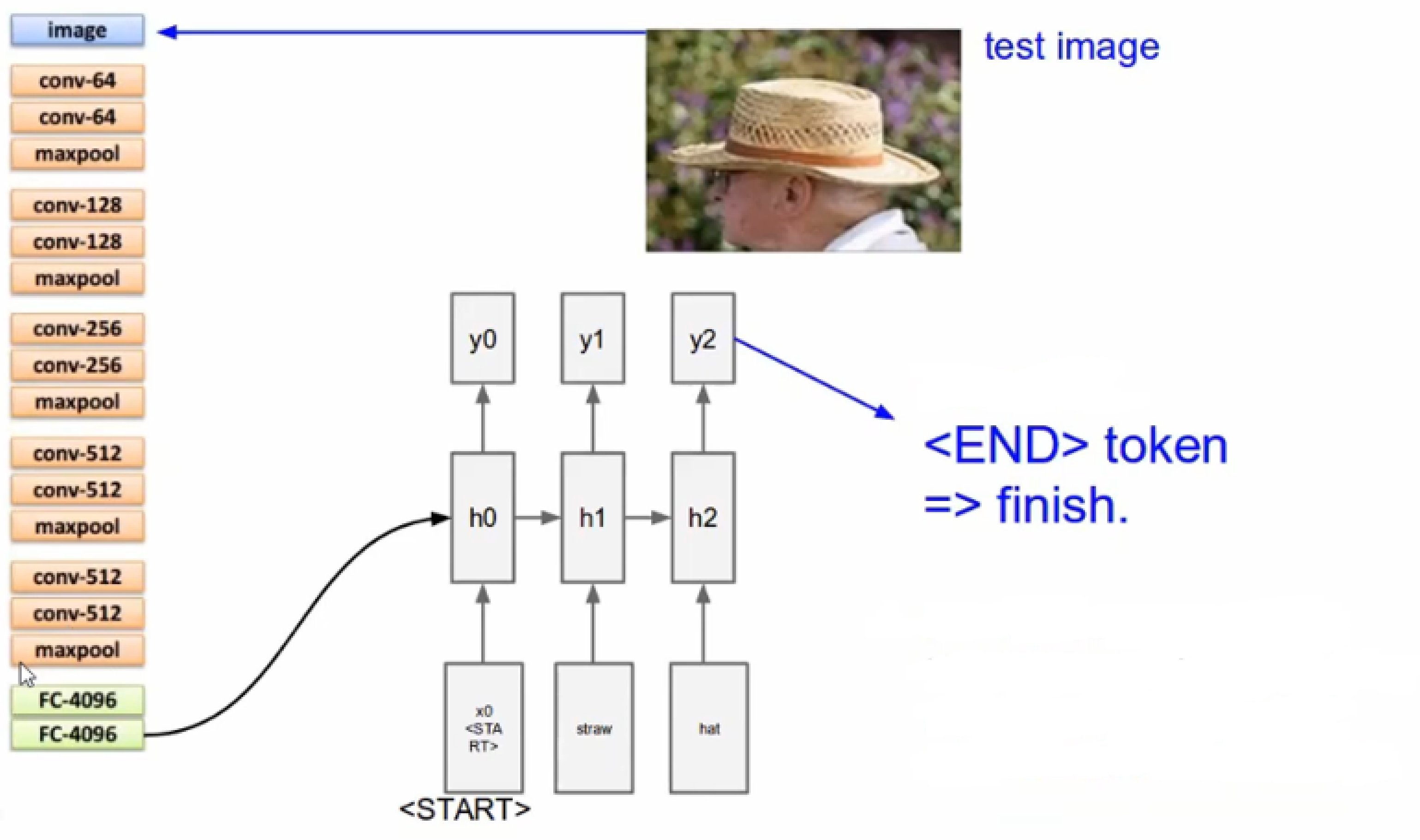
ВВЕДЕНИЕ

Задачи:

1. Обработать исходный набор данных для дальнейшего обучения и тестирования нейросетей;
2. Реализовать алгоритм текстового описания изображения на основе CNN и LSTM;
3. Реализовать алгоритм текстового описания изображения на основе CNN и Transformer decoder;
4. Реализовать алгоритм текстового описания изображения на основе CNN и Transformer;
5. Сравнить качество генерируемых описаний в каждом из подходов.

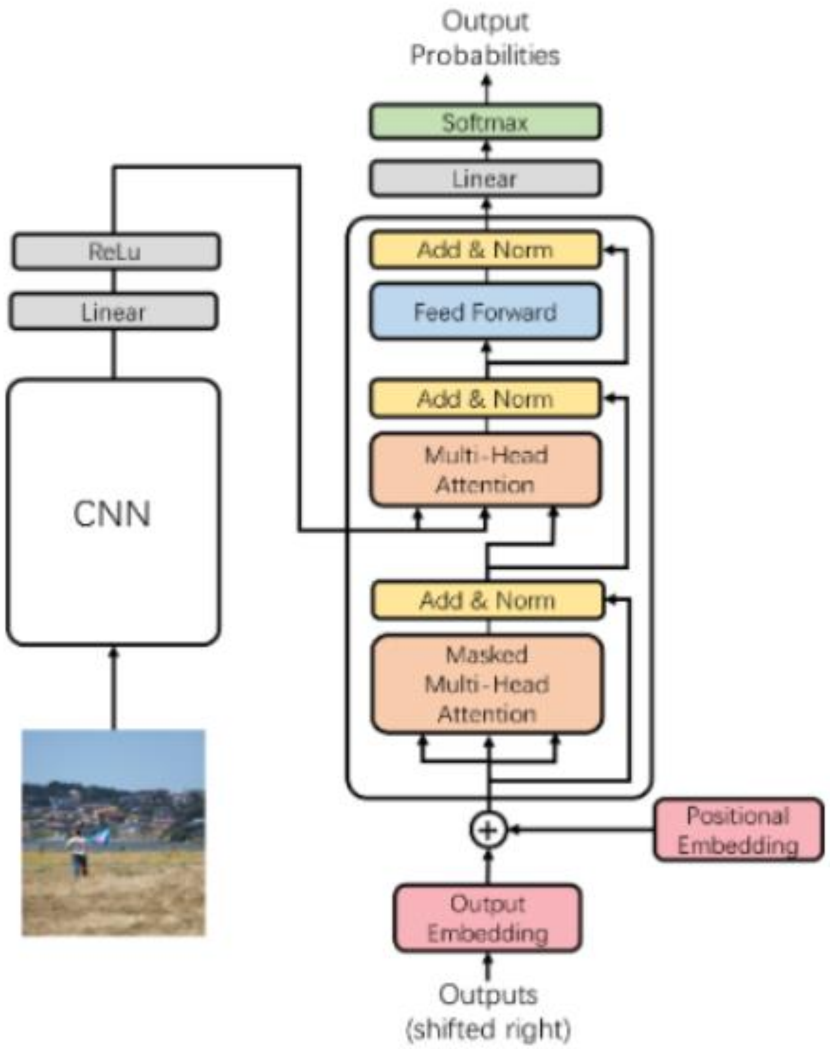
CNN + LSTM

ИНЖЕНЕРНАЯ ШКОЛА
ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ
И РОБОТОТЕХНИКИ



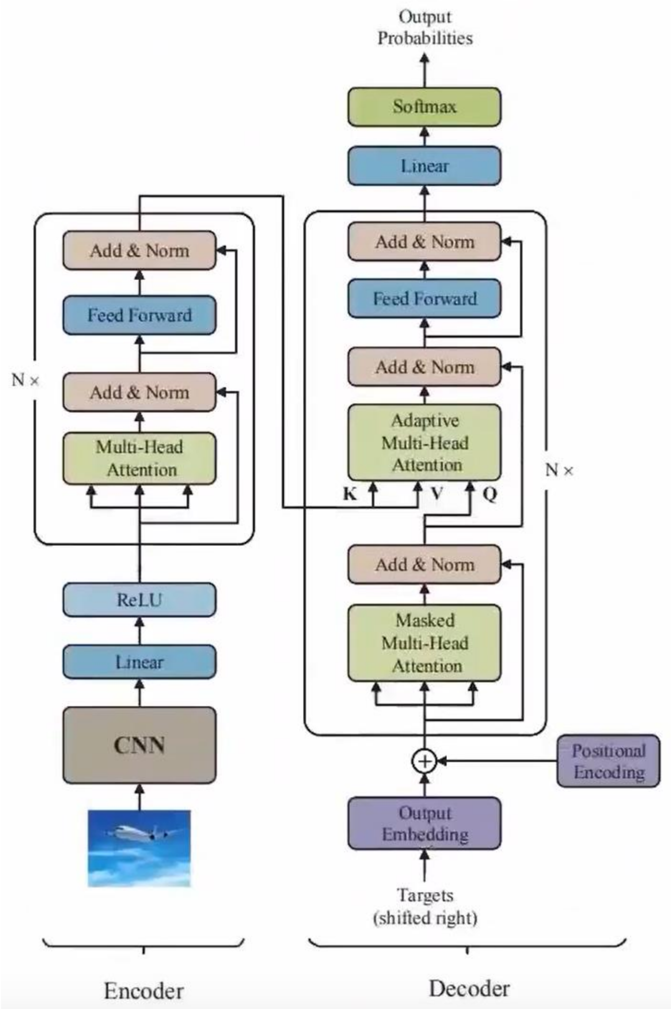
CNN + TRANSFORMER DECODER

ИНЖЕНЕРНАЯ ШКОЛА ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ И РОБОТОТЕХНИКИ

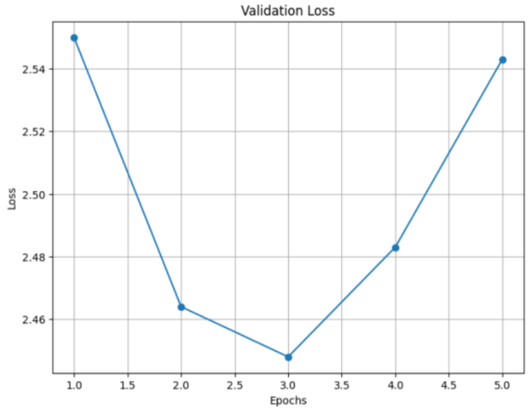
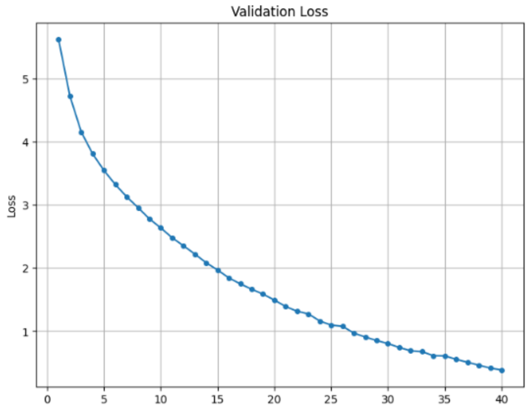
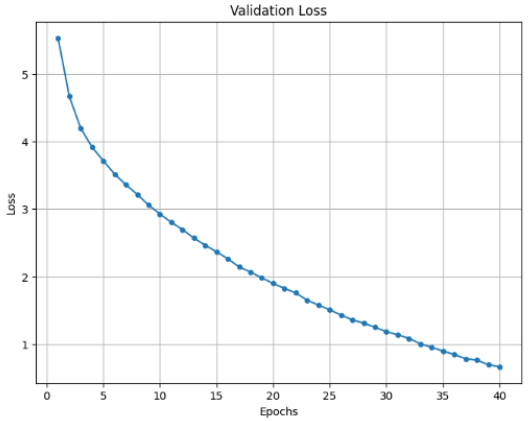
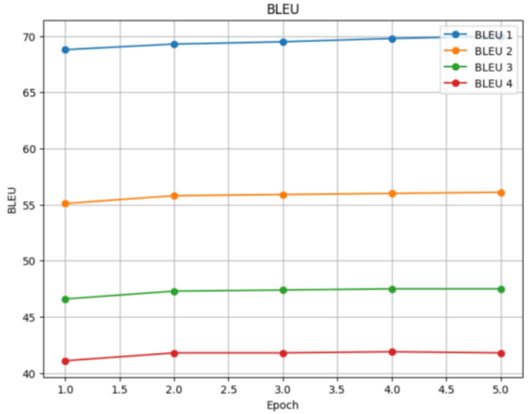
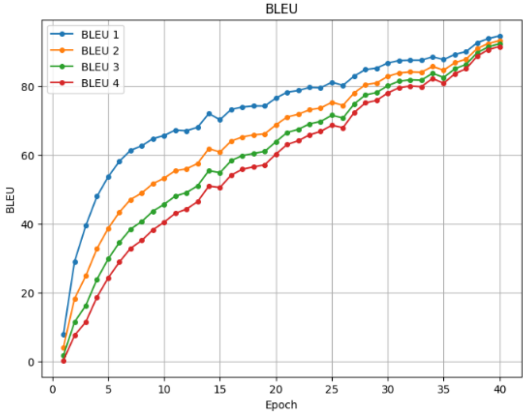
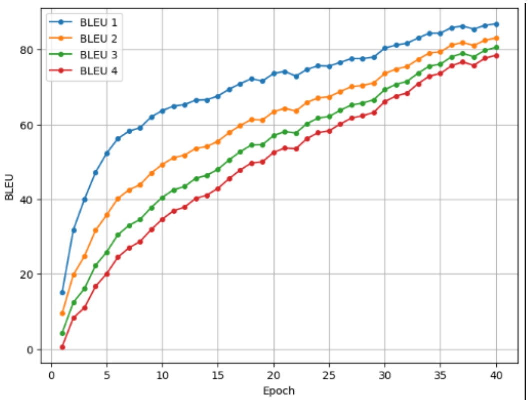
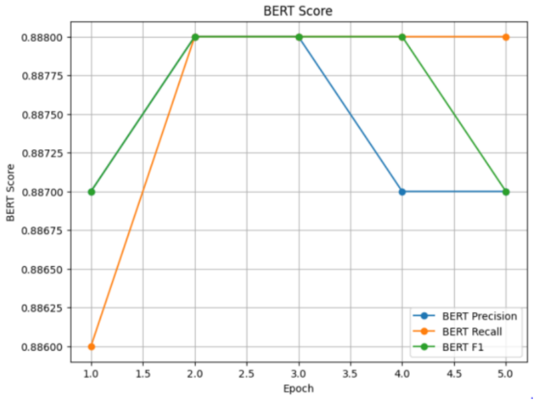
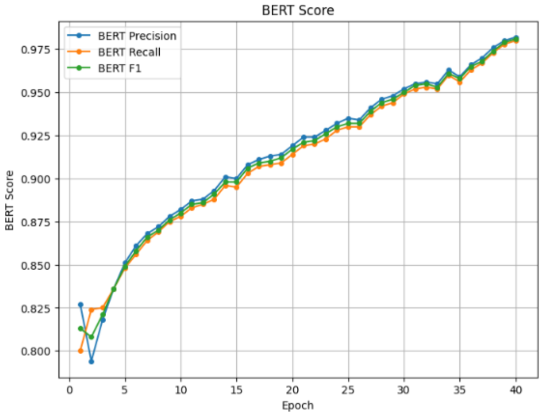
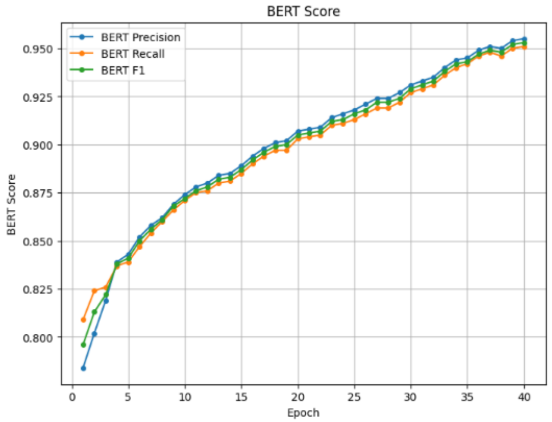


CNN + TRANSFORMER

ИНЖЕНЕРНАЯ ШКОЛА
ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ
И РОБОТОТЕХНИКИ



Метрики	VGG16+LSTM (seq_len = 8)	VGG16 + Transformer Decoder (seq_len = 8)	VGG16 + Transformer (seq_len = 8)
Val loss	2.543	0.385	0.664
BLEU 1	70.0	94.7	86.9
BLEU 2	56.1	93.3	83.1
BLEU 3	47.5	92.4	80.6
BLEU 4	41.8	91.6	78.5
BERT Precision	0.887	0.982	0.955
BERT Recall	0.888	0.98	0.951
BERT F1	0.887	0.981	0.953
Количество эпох	5	40	40
Время обучения, мин	25	10	11

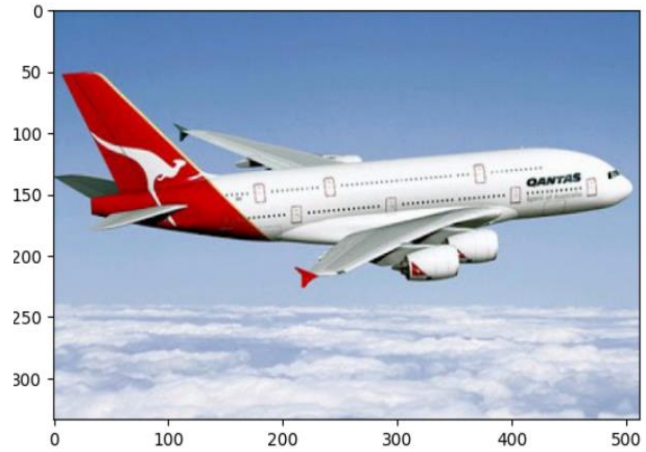
	VGG16+LSTM (seq_len = 8)	VGG16 + Transformer Decoder (seq_len = 8)	VGG16 + Transformer (seq_len = 8)
Validation loss			
BLEU			
BERT Score			

VGG16+LSTM (seq_len = 8)	VGG16 + Transformer Decoder (seq_len = 8)	VGG16 + Transformer (seq_len = 8)

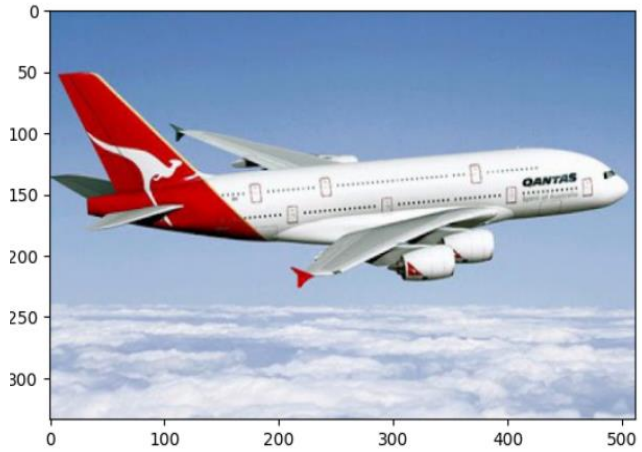
a man riding a bike on

a man riding a bike that

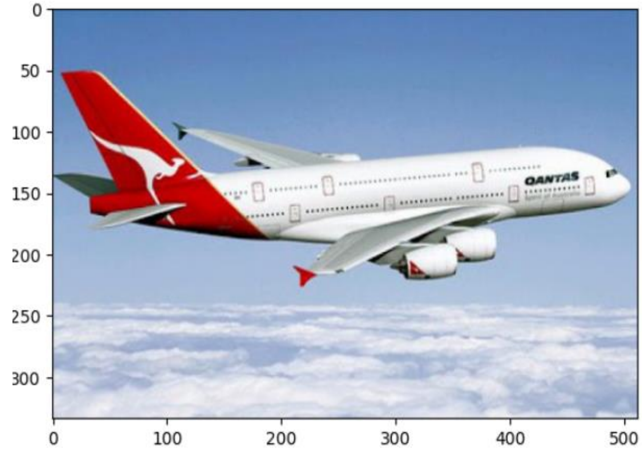
a man riding a bike through



a large passenger jet flying over



a large passenger jet on top



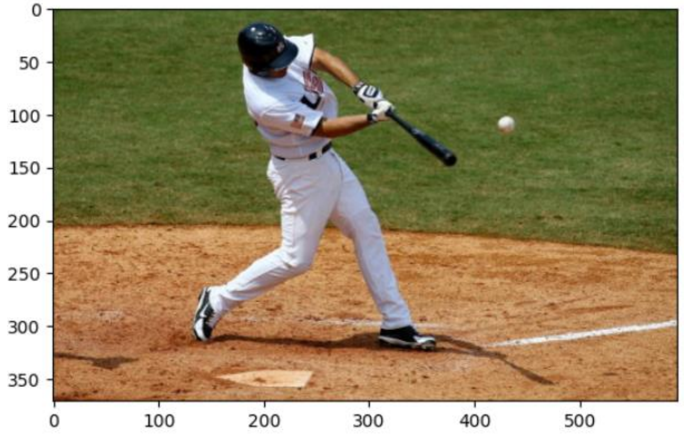
a large plane flying through the

VGG16+LSTM (seq_len = 8)	VGG16 + Transformer Decoder (seq_len = 8)	VGG16 + Transformer (seq_len = 8)

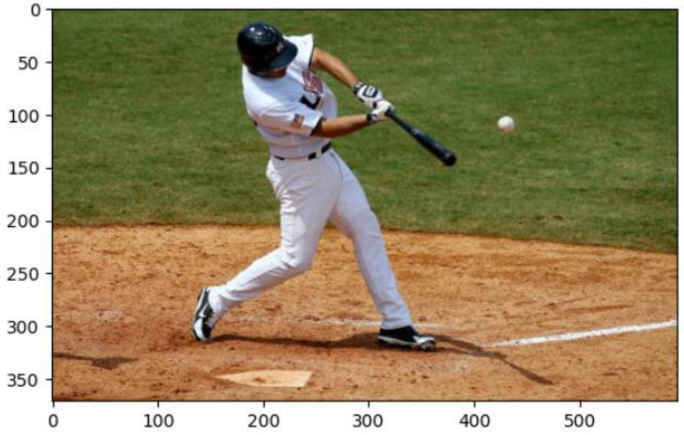
a truck with a large white

a bus car on the corner

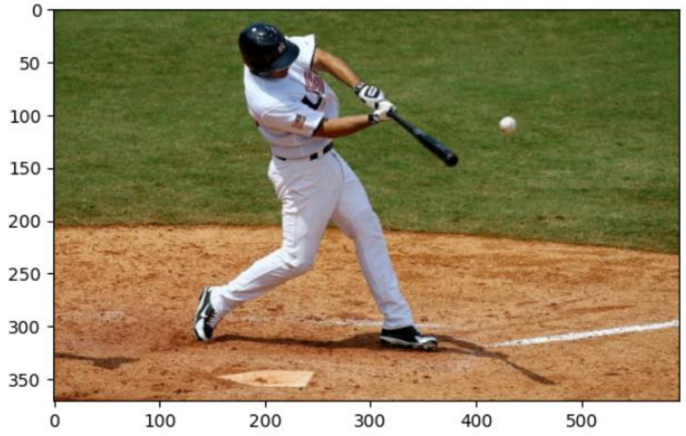
a small child in a field



a baseball player holding a baseball

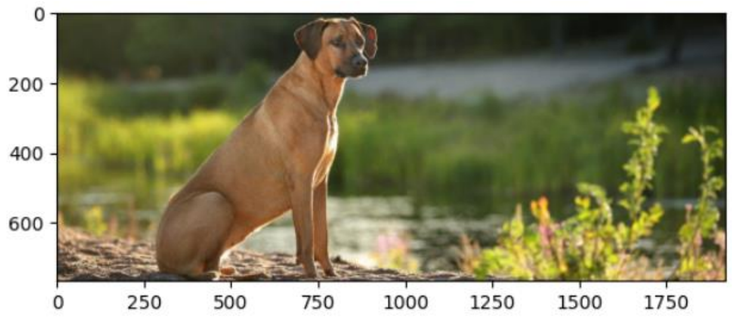


a baseball player holding a baseball

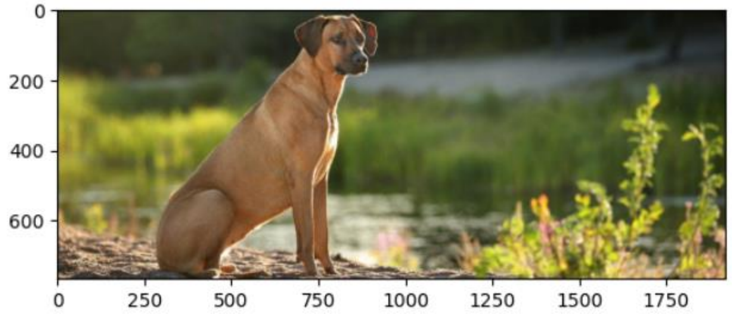


a baseball player holding a bat

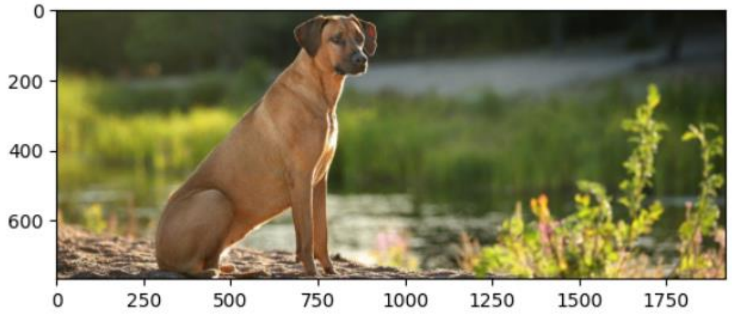
VGG16+LSTM (seq_len = 8)	VGG16 + Transformer Decoder (seq_len = 8)	VGG16 + Transformer (seq_len = 8)



a brown horse standing on a



a dog is standing in a



a brown dog holding a brown

a bird flying in the sky

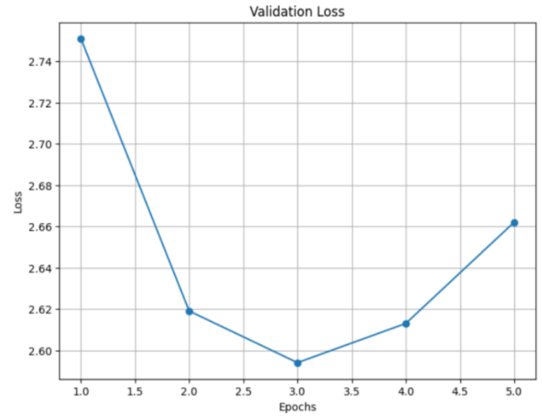
a large bird flying through the

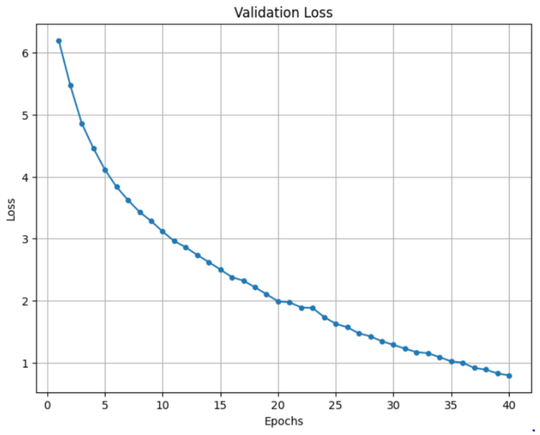
a bird standing in the air

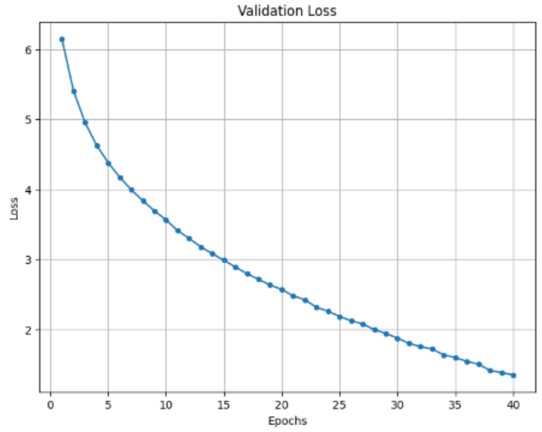
Метрики	VGG16+LSTM (seq_len = 12)	VGG16 + Transformer Decoder (seq_len = 12)	VGG16 + Transformer (seq_len = 12)
Val loss	2.662	0.794	1.347
BLEU 1	73.9	87.9	81.0
BLEU 2	59.8	83.7	73.1
BLEU 3	50.7	81.0	68.0
BLEU 4	44.5	78.8	64.2
BERT Precision	0.888	0.952	0.926
BERT Recall	0.892	0.949	0.923
BERT F1	0.89	0.951	0.925
Количество эпох	5	40	40
Время обучения, мин	61	14	16

VGG16+LSTM (seq_len = 12)	VGG16 + Transformer Decoder (seq_len = 12)	VGG16 + Transformer (seq_len = 12)

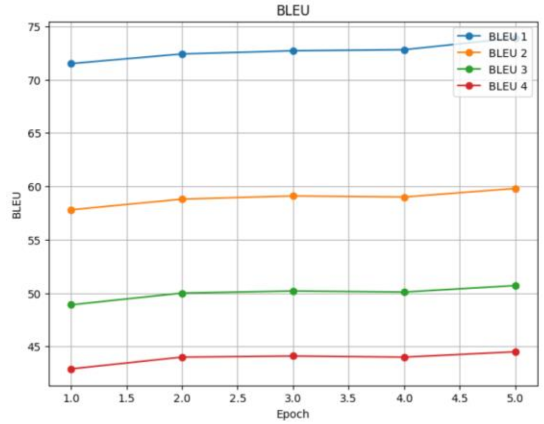
Validation loss

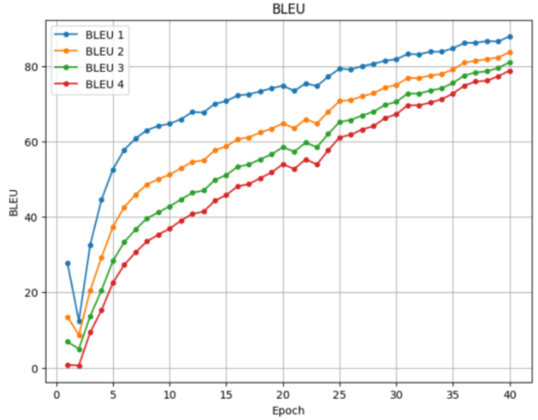


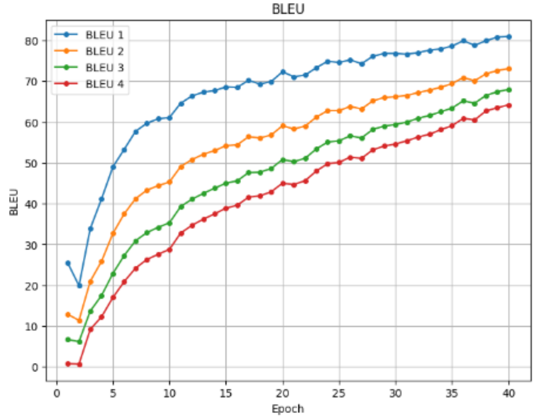




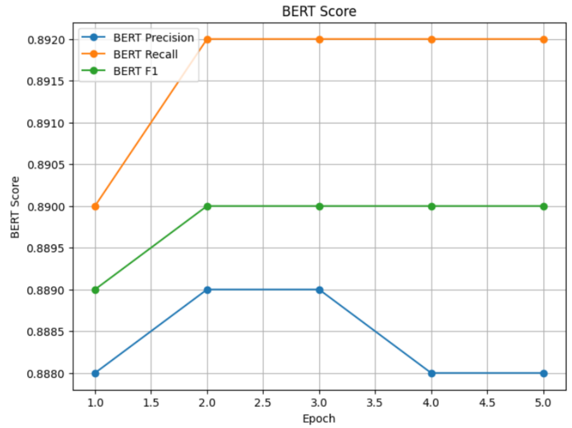
BLEU

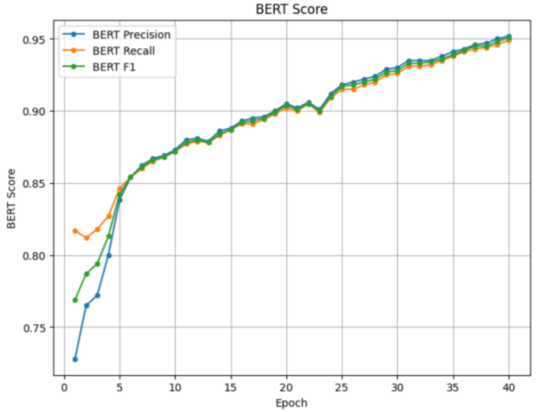


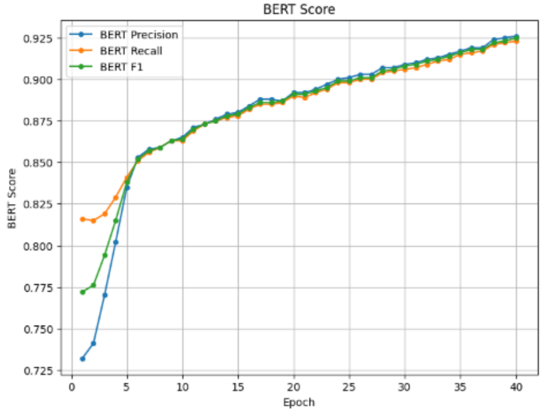


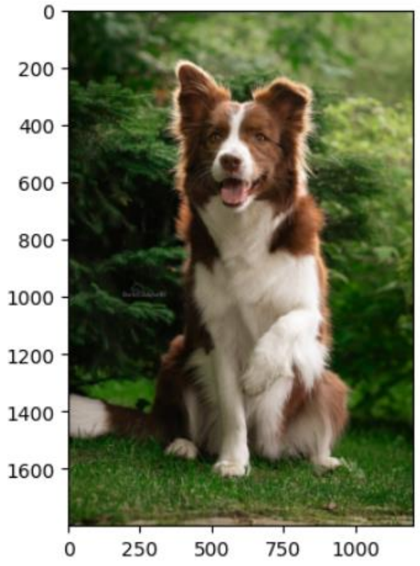
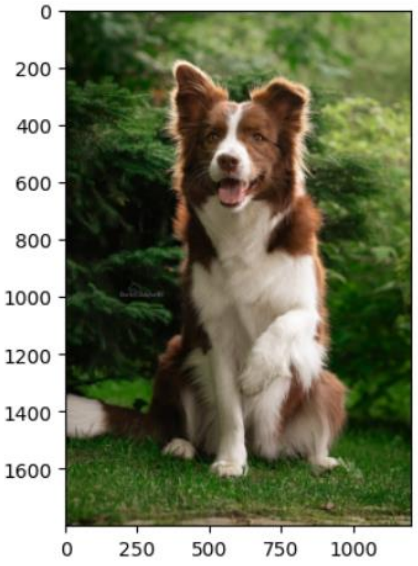
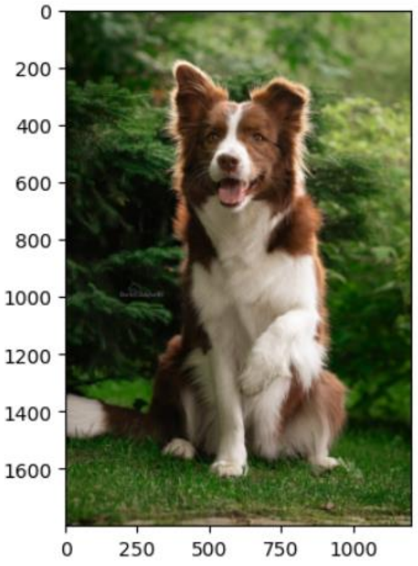


BERT Score

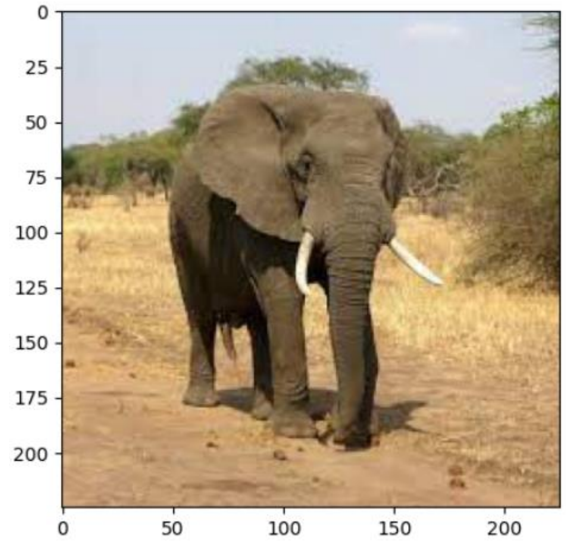
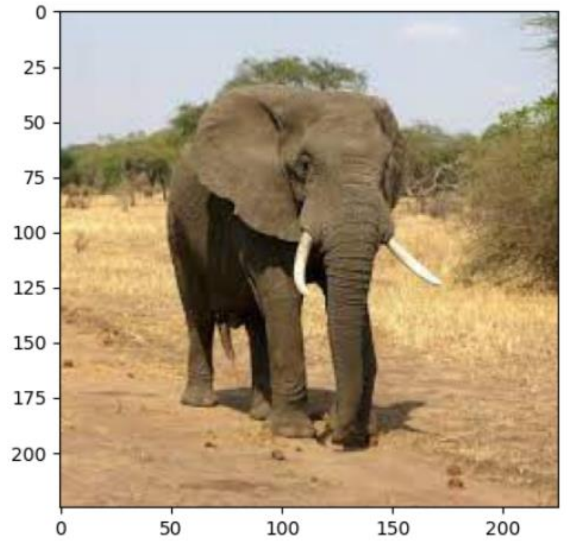
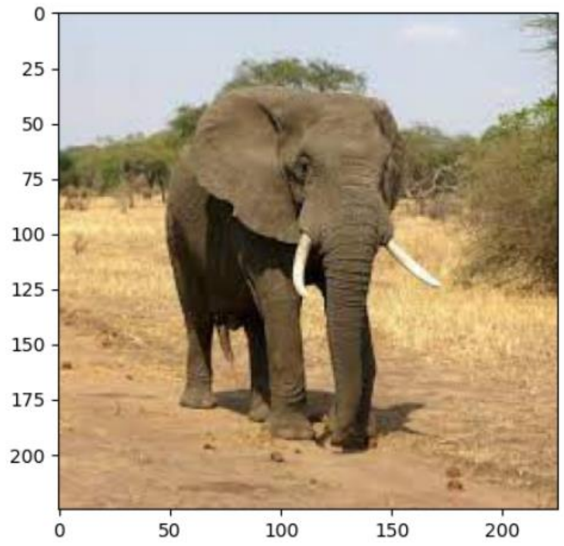






VGG16+LSTM (seq_len = 12)	VGG16 + Transformer Decoder (seq_len = 12)	VGG16 + Transformer (seq_len = 12)
		
a brown and white dog standing in a field .	a dog that is standing in the grass .	a dog laying in a brown dog with a red
		
a large passenger jet flying in the sky .	an airplane is flying through the air while flying over	a large plane flying through an airport runway.

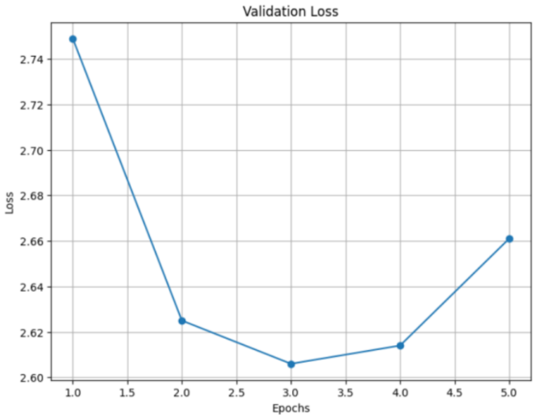
VGG16+LSTM (seq_len = 12)	VGG16 + Transformer Decoder (seq_len = 12)	VGG16 + Transformer (seq_len = 12)
		
a man riding a skateboard on a cement ramp .	a man wearing a skateboard trick on a skateboard .	a man riding a skateboard up a skateboard up a
		
a refrigerator with a bunch of food in it	a refrigerator and white plate filled with food .	a glass table with three glass vase of food .

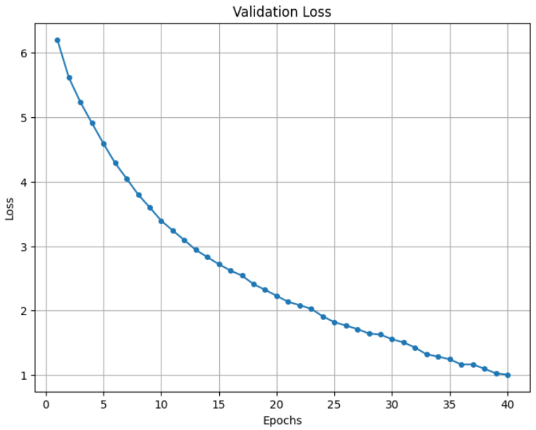
VGG16+LSTM (seq_len = 12)	VGG16 + Transformer Decoder (seq_len = 12)	VGG16 + Transformer (seq_len = 12)
		
a herd of elephants walking across a dirt road .	two elephants in some elephants standing on some trees .	two elephants in some elephants standing on some trees .
		
a man riding a snowboard down a snow covered slope	a man riding a snowboard on the side of a	a man riding skis down a snow covered slope .

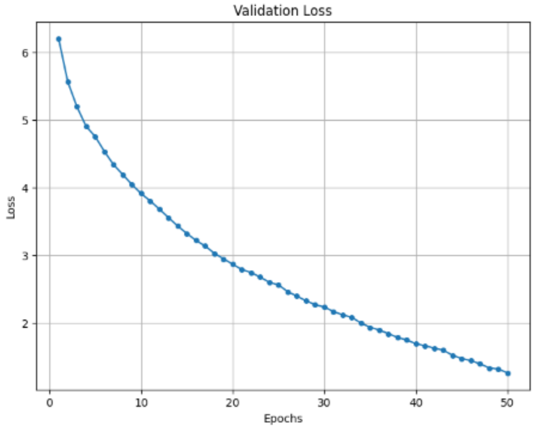
Метрики	VGG16+LSTM (seq_len = 16)	VGG16 + Transformer Decoder (seq_len = 16)	VGG16 + Transformer (seq_len = 16)
Val loss	2.661	1.005	1.264
BLEU 1	74.3	84.3	82.2
BLEU 2	60.1	78.6	74.4
BLEU 3	50.7	74.8	69.2
BLEU 4	44.3	71.8	65.4
BERT Precision	0.878	0.931	0.922
BERT Recall	0.895	0.936	0.927
BERT F1	0.887	0.933	0.924
Количество эпох	5	40	40
Время обучения, мин	48	17	22

VGG16+LSTM (seq_len = 16)	VGG16 + Transformer Decoder (seq_len = 16)	VGG16 + Transformer (seq_len = 16)

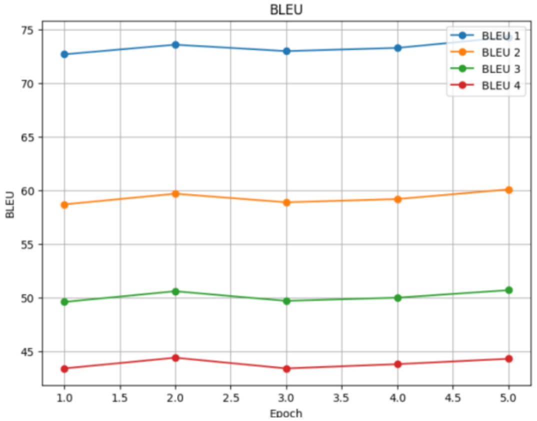
Validation loss

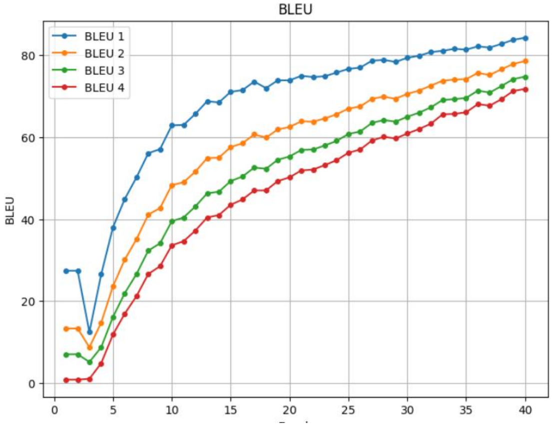


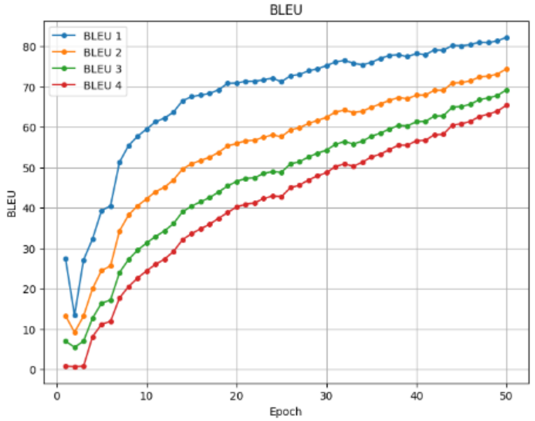




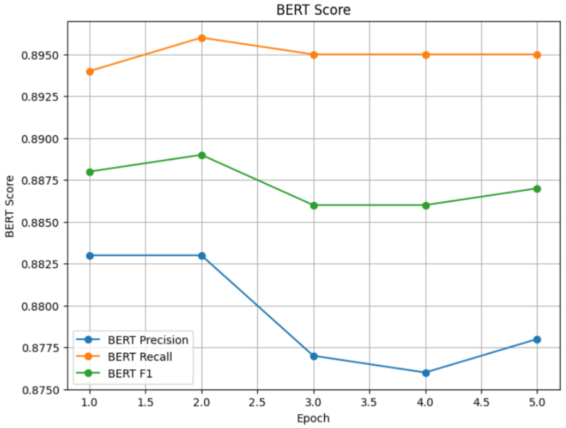
BLEU

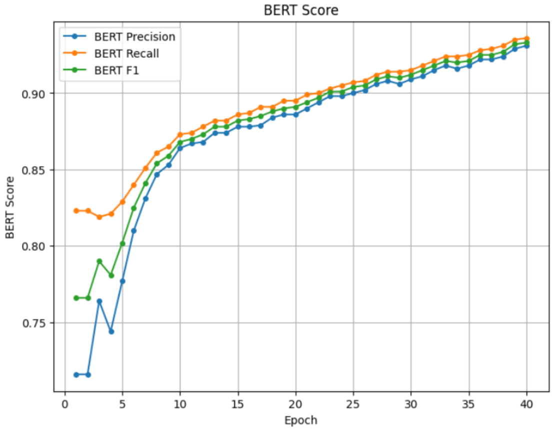


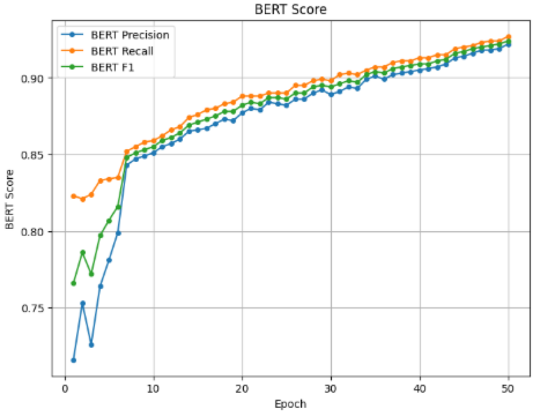




BERT Score







VGG16+LSTM (seq_len = 16)	VGG16 + Transformer Decoder (seq_len = 16)	VGG16 + Transformer (seq_len = 16)

a red and white car coll ##iding with a man on a motorcycle

a car parked next to a car covered car .

a car parked in front of a parking lot .

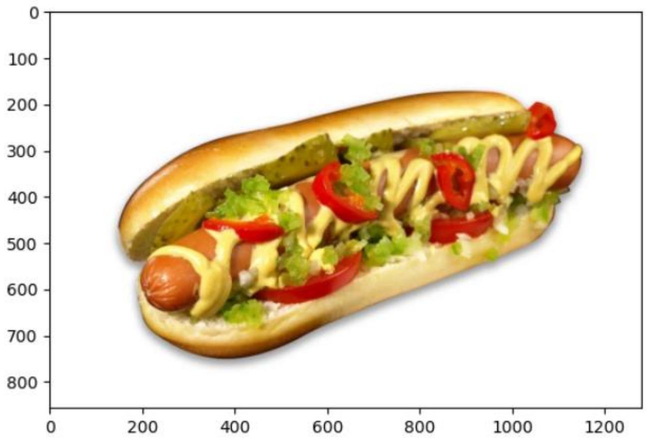
two horses standing on a lush green field .

a group of horses grazing next to a brown horse .

a man standing in front of a brown horse in the water .

VGG16+LSTM (seq_len = 16)	VGG16 + Transformer Decoder (seq_len = 16)	VGG16 + Transformer (seq_len = 16)

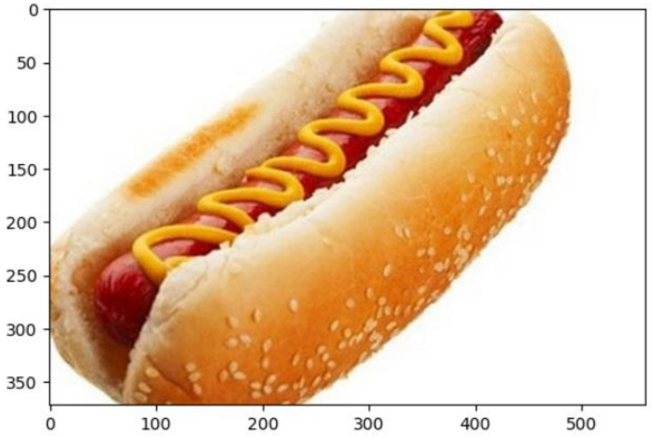
VGG16+LSTM (seq_len = 16)



a close up of a hot dog with ketchup and mustard

a man is looking at a stove with microwave .

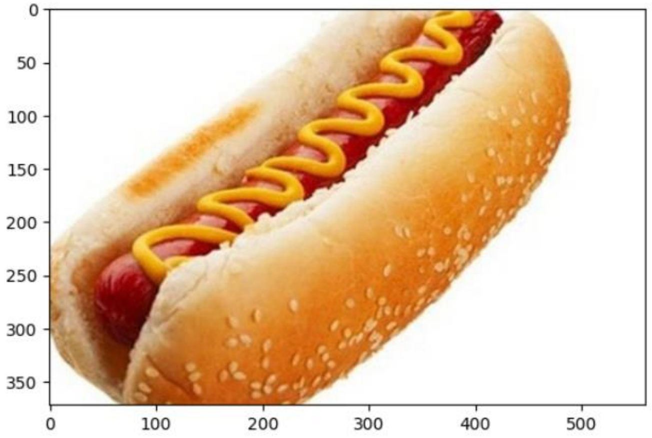
VGG16 + Transformer Decoder (seq_len = 16)



a hot dog on a plate with a plate .

a little girl with a stove , and holding a plate of food .

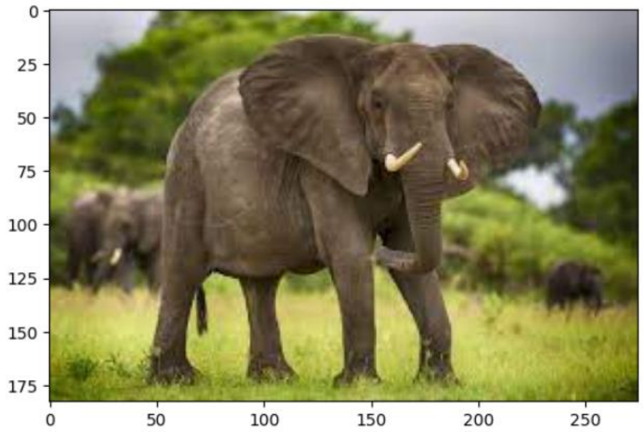
VGG16 + Transformer (seq_len = 16)



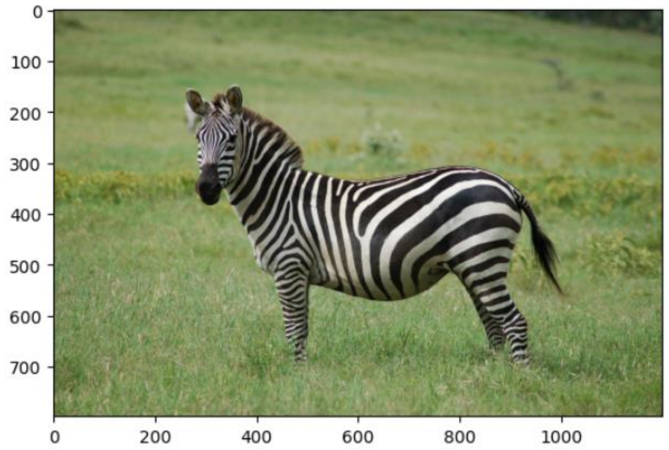
a hot dog in the middle of a bun .

a person holding a pizza in front of a microwave .

VGG16+LSTM (seq_len = 16)

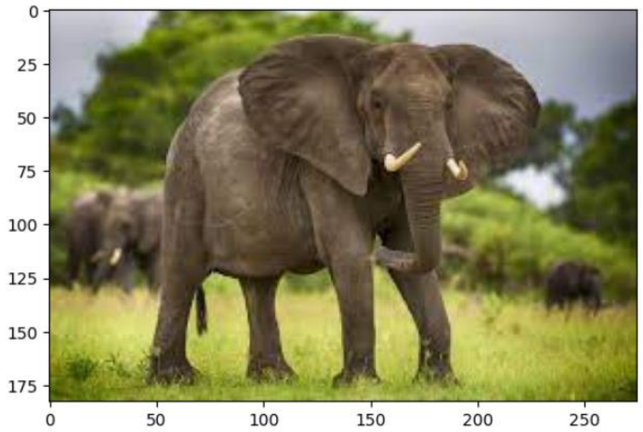


a large elephant standing in a field with a baby elephant nearby .

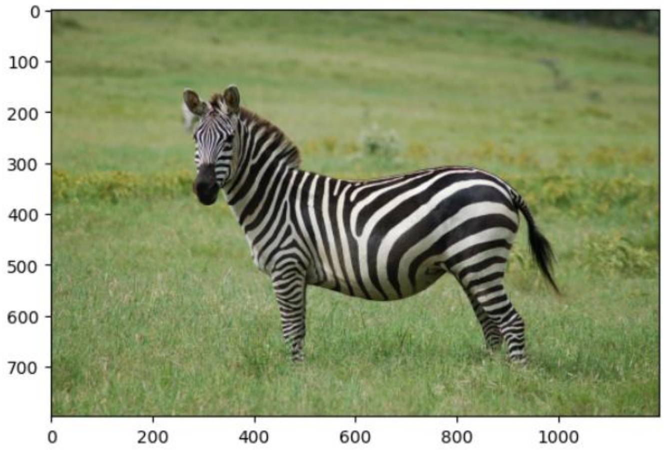


a zebra standing on a lush green field .

VGG16 + Transformer Decoder (seq_len = 16)

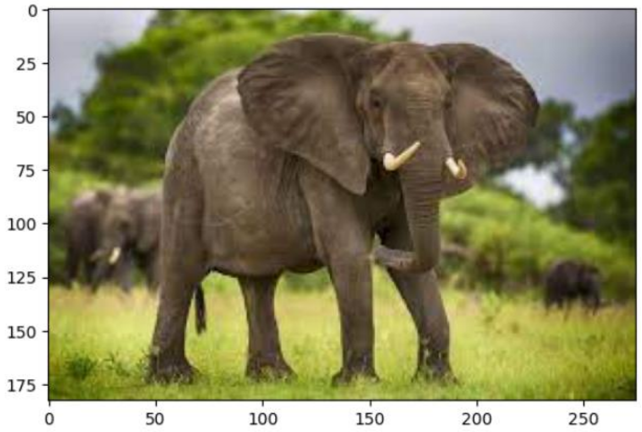


a baby elephant walking on a green field .

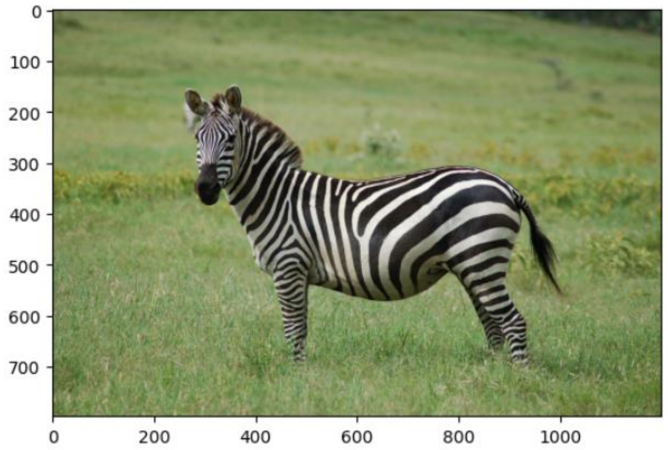


a zebra standing in a field near grass field

VGG16 + Transformer (seq_len = 16)



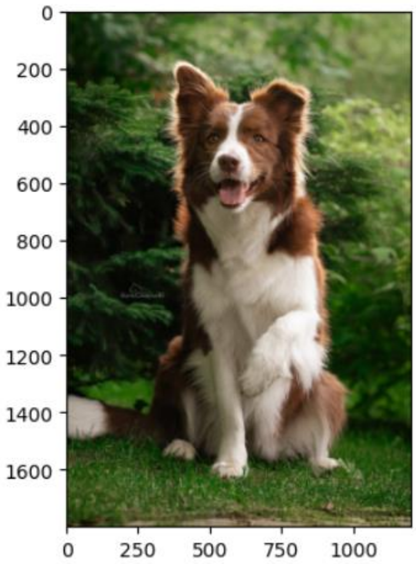
a couple of elephants standing next to each other



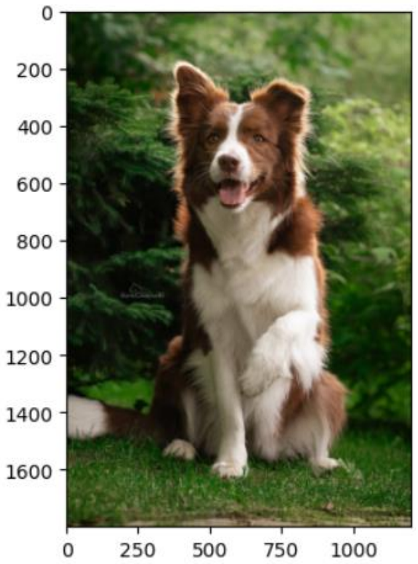
a zebra standing in front of a zebra standing next to each other

Метрики	Efficientnet_v2_s + Transformer Decoder (seq_len = 12)	VGG16 + Transformer Decoder (seq_len = 12)	ResNet18 + Transformer Decoder (seq_len = 12)
Val loss	1.669	0.794	1.894
BLEU 1	77.5	87.9	75.5
BLEU 2	67.4	83.7	64.3
BLEU 3	60.7	81.0	56.7
BLEU 4	56.0	78.8	51.4
BERT Precision	0.909	0.952	0.9
BERT Recall	0.909	0.949	0.901
BERT F1	0.909	0.951	0.9
Количество эпох	40	40	40
Время обучения, мин	14	14	13.9

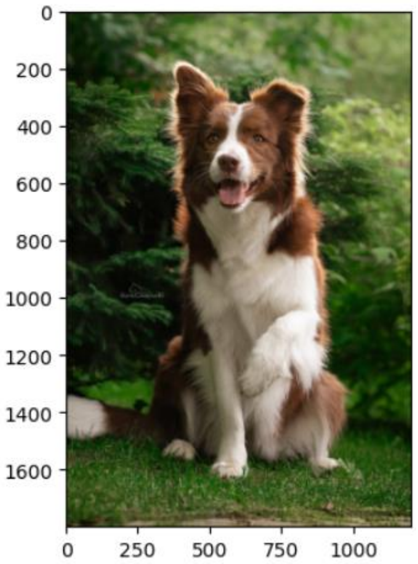
Efficientnet_v2_s + Transformer Decoder (seq_len = 12)	VGG16 + Transformer Decoder (seq_len = 12)	ResNet18 + Transformer Decoder (seq_len = 12)



a cat are player that are player ##ing head out



a dog that is standing in the grass .



a man standing on a tennis court holding a racquet

a city street with a city street at a street

an airplane is flying through the air while flying over

a man is riding a wave on top of a

Метрики	VGG16 + Transformer Decoder (seq_len = 12, n_head = 8)	VGG16 + Transformer Decoder (seq_len = 12, n_head = 16)	VGG16 + Transformer Decoder (seq_len = 12, n_head = 64)
Val loss	1.238	1.194	1.152
BLEU 1	79.9	80.5	81.8
BLEU 2	73.3	74.0	75.6
BLEU 3	68.89	69.7	71.5
BLEU 4	65.5	66.3	68.2
BERT Precision	0.925	0.926	0.93
BERT Recall	0.922	0.923	0.927
BERT F1	0.923	0.925	0.929
Количество эпох	40	40	40
Время обучения, мин	13.36	13.41	17.16

VGG16 + Transformer Decoder (seq_len = 12, n_head = 8)	VGG16 + Transformer Decoder (seq_len = 12, n_head = 16)	VGG16 + Transformer Decoder (seq_len = 12, n_head = 64)

a dog standing in a dog with a dog in

a dog with a dog and a brown in

a dog and dog standing in front of a dog

a kitchen with a pile of food in a kitchen

a kitchen with a refrigerator and a refrigerator .

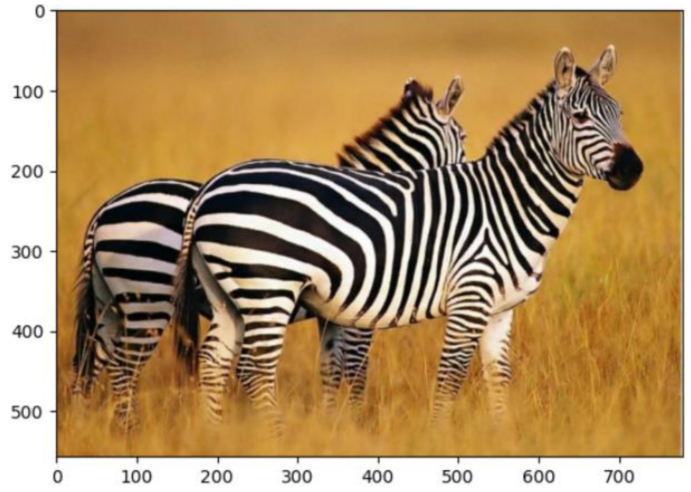
a kitchen filled with lots of food in a kitchen

VGG16 + Transformer Decoder (seq_len = 12, n_head = 8)	VGG16 + Transformer Decoder (seq_len = 12, n_head = 16)	VGG16 + Transformer Decoder (seq_len = 12, n_head = 64)

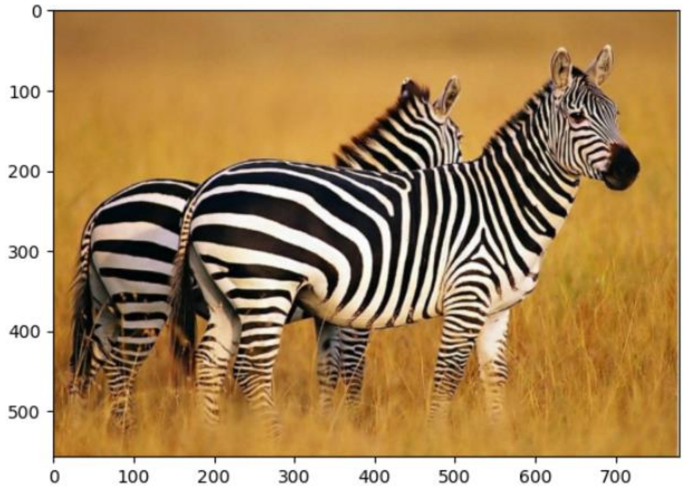
a man riding a snowboard on top of a snowboard

a man riding skis on top of a snowboard .

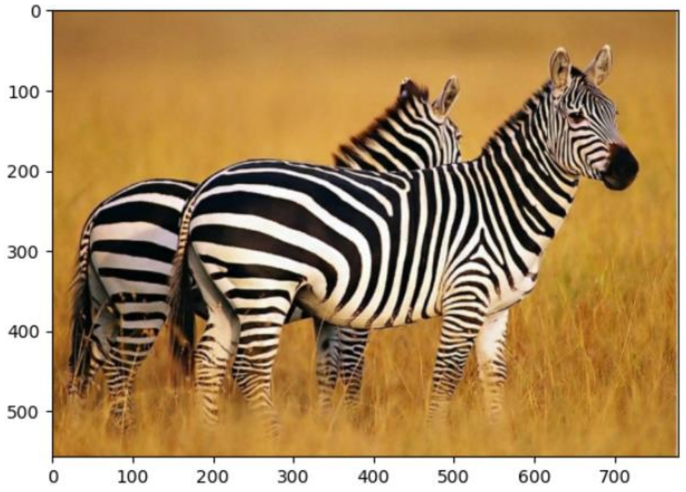
a man flying through a snowboard on top of a



a group of zebra standing in a grassy area .



a zebra standing in the grass with a zebra standing



a group of zebra standing around a zebra standing in

ЗАКЛЮЧЕНИЕ

В ходе выполнения выпускной квалификационной работы были рассмотрены, реализованы и сравнены между собой три нейросетевых архитектуры моделей для генерации текстовых описаний изображения.

Результаты экспериментов показали, что наилучшие показатели по метрикам BLEU, BERT score и визуальной оценке были достигнуты моделью, использующей VGG16 в качестве визуального энкодера, и Transformer Decoder с 8 головами внимания.

Данная архитектура продемонстрировала способность генерировать наиболее содержательные, связные и релевантные описания изображений по сравнению с другими рассмотренными подходами, при наименьшем времени обучения.

Был проведен анализ финансовой и ресурсной эффективности рассматриваемого проекта, который показал, что наиболее конкурентоспособной и эффективной в коммерциализации является та же модель VGG16+Transformer Decoder.

Также были выявлены возможные вредные и опасные факторы, которые могут возникнуть во время эксплуатации продукта, и пути их устранения.

CONCLUSION

During the completion of the final qualifying work, three neural network model architectures for generating textual descriptions of images were considered, implemented, and compared.

The experimental results showed that the best performance in terms of BLEU, BERT score, and visual evaluation metrics was achieved by the model using VGG16 as the visual encoder and Transformer Decoder with 8 attention heads.

This architecture demonstrated the ability to generate the most informative and relevant image descriptions compared to the other approaches considered, with the shortest training time.

A financial and resource efficiency analysis of the project was conducted, which showed that the same VGG16+Transformer Decoder model is the most competitive and commercially viable.

Potential harmful and dangerous factors that may arise during product operation, as well as ways to eliminate them, have also been identified.

ТОМСКИЙ ПОЛИТЕХНИЧЕСКИЙ УНИВЕРСИТЕТ

ИНЖЕНЕРНАЯ ШКОЛА ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ И РОБОТОТЕХНИКИ

РАЗРАБОТКА АЛГОРИТМА ГЕНЕРАЦИИ ТЕКСТОВОГО ОПИСАНИЯ ИЗОБРАЖЕНИЯ НА ОСНОВЕ НЕЙРОСЕТЕВЫХ АРХИТЕКТУР

Выпускная квалификационная работа

Научный руководитель: Спицын В.Г., д.т.н., профессор ОИТ ТПУ

Консультант: Кривошеев Н.А., старший преподаватель ОИТ ТПУ

Выполнил: Ямкин Н. Н.

Группа: 8ВМ22

20.06.2024